Solution:

Computation of trend values by the method of least squares.

In case of EVEN number of years, let us consider

$$X = \frac{(x - Arithimetic\ mean\ of\ two\ middle\ years)}{0.5}$$

Year (x)	Sales (Y)	$X = \frac{(x - 1998.5)}{0.5}$	XY	X^2	Trend values (Y_t)
1995	6.7	–7	–46.9	49	5.6167
1996	5.3	–5	–26.5	25	5.7190
1997	4.3	–3	–12.9	9	5.8214
1998	6.1	–1	–6.1	1	5.9238
1999	5.6	1	5.6	1	6.0262
2000	7.9	3	23.7	9	6.1286
2001	5.8	5	29.0	25	6.2310
2002	6.1	7	42.7	49	6.3333
N=8	47.8	$\Sigma X = 0$	8.6	168	

Table 9.6

$$a = \frac{\sum Y}{n} = \frac{47.8}{8} = 5.975$$

$$b = \frac{\sum XY}{\sum X^2} = \frac{8.6}{168} = 0.05119$$

Therefore, the required equation of the straight line trend is given by

$Y = a + bX$; $Y = 5.975 + 0.05119\,X$.

When $X = 1995$,

$$X = 1995,\ Y_t = 5.975 + 0.05119\left(\frac{1995 - 1998.5}{0.5}\right) = 5.6167$$

When $X = 1996$,

$$X = 1996,\ Y_t = 5.975 + 0.05119\left(\frac{1996 - 1998.5}{0.5}\right) = 5.7190$$

similarly other values can be obtained.

Note

(i) Future forecasts made by this method are based only on trend values.
(ii) The predicted values are more reliable in this method than the other methods.

9.1.5 Methods of measuring Seasonal Variations By Simple Averages :

Seasonal Variations can be measured by the method of simple average. The data should be available in season wise likely weeks, months, quarters.

Method of Simple Averages:

This is the simplest and easiest method for studying Seasonal Variations. The procedure of simple average method is outlined below.

Procedure:

(i) Arrange the data by months, quarters or years according to the data given.

(ii) Find the sum of the each months, quarters or year.

(iii) Find the average of each months, quarters or year.

(iv) Find the average of averages, and it is called Grand Average (G)

(v) Compute Seasonal Index for every season (i.e) months, quarters or year is given by

$$\text{Seasonal Index (S.I)} = \frac{Seasonal\ Average}{Grand\ average} \times 100$$

(vi) If the data is given in months

$$\text{S.I for Jan} = \frac{Monthly\ Average\ (for\ Jan)}{Grand\ average} \times 100$$

$$\text{S.I for Feb} = \frac{Monthly\ Average\ (for\ Feb)}{Grand\ average} \times 100$$

Similarly we can calculate SI for all other months.

(vii) If the data is given in quarter

$$\text{S.I for I Quarter} = \frac{Average\ of\ I\ quarter}{Grand\ average} \times 100$$

$$\text{S.I for II Quarter} = \frac{Average\ of\ II\ quarter}{Grand\ average} \times 100$$

$$\text{S.I for III Quarter} = \frac{Average\ of\ III\ quarter}{Grand\ average} \times 100$$

$$\text{S.I for IV Quarter} = \frac{Average\ of\ IV\ quarter}{Grand\ average} \times 100$$

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